

1.7.13: Summary

This TLP has covered the following topics:

- The mathematical description of the atomic structure of a crystal.
- The different lattice types.
- What a lattice vector, unit cell & motif are and how they are relevant to describing crystals.
- Close packing and the packing efficiency of a crystal structure.
- What symmetry operators are present within a crystallographic lattice and how only specific combinations of such symmetry elements may exist.
- Bravais lattice and Point Groups.
- How to construct the plan view of a 2×2 unit cell.

Going further

Books

- A. Putnis, *Introduction to Mineral Sciences*, Cambridge University Press, 1992
- Feynman, Leighton, Sands, *Lectures on Physics Vol. II, Chapter 30*, Addison-Wesley Publishing Company, 1964 [Note: This book follows a different convention when referring to body-centred and face-centred structures]
- C. Hammond, *The Basics of Crystallography and Crystallography*, III Ed, Oxford University Press, 2009
- B.D Cullity, S.R Stock, *Elements of X-Ray Diffraction*, Prentice Hall, 2001
- McKie, McKie, *Essentials of Crystallography*, Blackwell Scientific Publications, 1986

Website of **International Tables for Crystallography** <http://it.iucr.org/A/>

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